

$$\textcircled{1} f(0,0) = 1$$

$$\frac{\partial f}{\partial x} = \frac{1}{2(1-x+2y^2)^{3/2}} \dots \frac{\partial f}{\partial x}(0,0) = \frac{1}{2}$$

$$\frac{\partial f}{\partial y} = -\frac{2y}{(1-x+2y^2)^{3/2}} \dots \frac{\partial f}{\partial y}(0,0) = 0$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{3}{4(1-x+2y^2)^{5/2}} \dots \frac{\partial^2 f}{\partial x^2}(0,0) = \frac{3}{4}$$

$$\frac{\partial^2 f}{\partial x \partial y} = -\frac{3y}{(1-x+2y^2)^{5/2}} \dots \frac{\partial^2 f}{\partial x \partial y}(0,0) = 0$$

$$\frac{\partial^2 f}{\partial y^2} = -\frac{2(1-x+2y^2)^{3/2} - 3y(1-x+2y^2)^{1/2} \cdot 4y}{(1-x+2y^2)^3}$$

$$\hookrightarrow \frac{\partial^2 f}{\partial y^2}(0,0) = -2$$

$$\Rightarrow T_2(x,y) = 1 + \frac{1}{2}x + \frac{3}{8}x^2 - y^2$$

$$f\left(\frac{1}{10}, \frac{1}{10}\right) \approx 1 + \frac{1}{20} + \left(\frac{3}{8} - 1\right) \frac{1}{100}$$

$$= 1 + \frac{1}{20} - \frac{5}{8} \frac{1}{100} = 1 + \frac{8-1}{160}$$

$$= \frac{167}{160}$$

$$\textcircled{2} \quad A = \begin{pmatrix} -1 & 1 \\ 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 2 \\ 2 \\ 3 \end{pmatrix}$$

a) Minimalizující funkce

$$f(x, y) = \left\| A \begin{pmatrix} x \\ y \end{pmatrix} - b \right\|^2 \quad \text{na } M = \mathbb{R}^2$$

$$b) \quad (x, y) \text{ je bod minima} \Leftrightarrow A^T A \begin{pmatrix} x \\ y \end{pmatrix} = A^T b$$

$$A^T A = \begin{pmatrix} -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 6 & 2 \\ 2 & 4 \end{pmatrix}$$

$$(A^T A)^{-1} = \frac{1}{20} \begin{pmatrix} 4 & -2 \\ -2 & 6 \end{pmatrix}$$

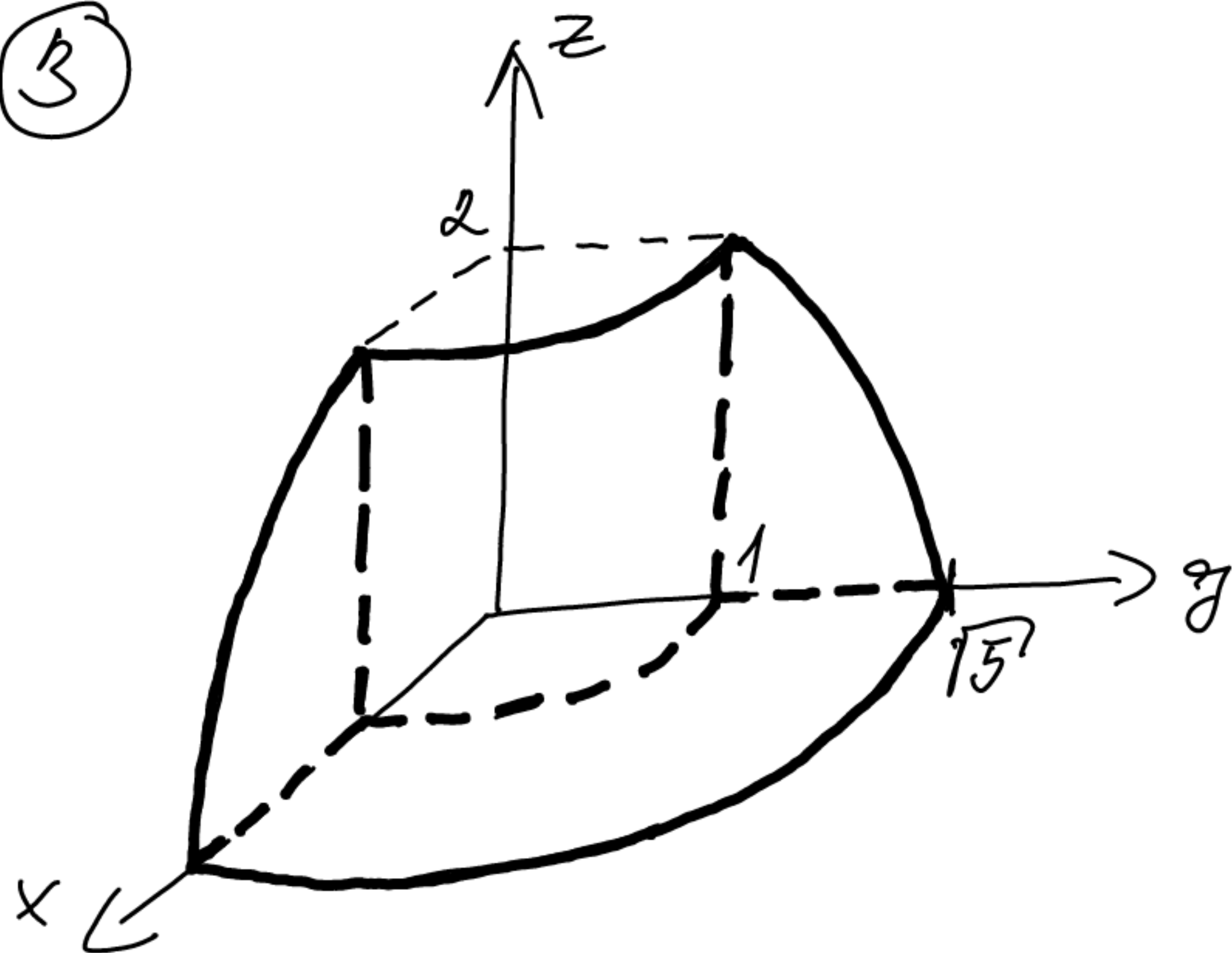
$$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{20} \begin{pmatrix} 4 & -2 \\ -2 & 6 \end{pmatrix} \begin{pmatrix} -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 2 \\ 2 \\ 3 \end{pmatrix}$$

$$= \frac{1}{10} \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 8 \\ 7 \end{pmatrix} = \frac{1}{10} \begin{pmatrix} 9 \\ 13 \end{pmatrix}$$

Hledaná přímka má rovnici

$$y = \frac{9}{10}x + \frac{13}{10}$$

3



$$\text{moment}(M) = \int_M x^2 + y^2 \, dV = \int_0^2 \int_0^{\frac{\pi}{2}} \int_1^{\sqrt{5-z^2}} r^3 \, dr \, d\theta \, dz$$

$$= \frac{\pi}{2} \int_0^2 \left[\frac{r^4}{4} \right]_1^{\sqrt{5-z^2}} dz = \frac{\pi}{8} \int_0^2 (5-z^2)^2 - 1 \, dz$$

$$= \frac{\pi}{8} \int_0^2 (24 - 10z^2 + z^4) \, dz$$

$$= \frac{\pi}{8} \left[24z - \frac{10}{3}z^3 + \frac{z^5}{5} \right]_0^2 = \pi \left(6 - \frac{10}{3} + \frac{4}{5} \right)$$

$$= \pi \frac{90 - 50 + 12}{15} = \frac{52\pi}{15}$$

④ Ad^c $F(x, y) = (0, x)$. Pal (Greenova
veta)

$$\text{obslah}(\Pi) = \int_{\Pi} 1 d\lambda_2 = \int_C \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} d\lambda_2 =$$

$$= \int_C F(x, y) \cdot ds = \int_0^{\pi} (0, \sin t \cos t) \cdot$$

$$= \int_0^{\pi} (0, \sin t \cos t) \cdot (\cos^2 t - \sin^2 t, \cos t) dt$$

$$= \int_0^{\pi} \sin t \cos^2 t dt = \left[-\frac{\cos^3 t}{3} \right]_0^{\pi}$$

$$= -\frac{1}{3}(-1 - 1) = \frac{2}{3}$$

$$\textcircled{5} \quad \lim_{\ell \rightarrow \infty} \frac{\ell}{4} \sqrt{\frac{(-1)^\ell (2\ell+1)}{4^\ell} x^{2\ell}} = \lim_{\ell \rightarrow \infty} \frac{\sqrt{2\ell+1}}{4} |x|^{2\ell}$$

$$= \frac{|x|^2}{4} < 1 \Leftrightarrow |x| < 2$$

Робота на конвергенца је $R=2$.

$$\sum_{\ell=1}^{\infty} \frac{(-1)^\ell (2\ell+1)}{4^\ell} x^{2\ell} = \left(\sum_{\ell=1}^{\infty} \frac{(-1)^\ell}{4^\ell} x^{2\ell+1} \right)'$$

$$= \left(x \sum_{\ell=1}^{\infty} \left(\frac{-x^2}{4} \right)^\ell \right)' = \left(-\frac{x^3}{4} \sum_{\ell=0}^{\infty} \left(\frac{-x^2}{4} \right)^\ell \right)'$$

$$= \left(-\frac{x^3}{4} \frac{1}{1 + \frac{x^2}{4}} \right)' = \left(-\frac{x^3}{4+x^2} \right)'$$

$$= -\frac{3x^2(4+x^2) - 2x^4}{(4+x^2)^2} = -\frac{12x^2 + x^4}{(4+x^2)^2}$$

по $|x| < 2$.